Teaching Notes by Course Objectives – Introduction to Machine Learning

CO1: Understand the concept of machine learning, types of learning, and applications.

Key Topics:

- What is Machine Learning (ML)?
- Difference between AI, ML, and Deep Learning
- Types of Machine Learning:■ - Supervised■ - Unsupervised■ - Semi-supervised■ - Reinforcement Learning
- Real-world Applications (e.g., spam detection, recommendation systems, medical diagnosis)

Teaching Aids:

- Use videos or animations to differentiate ML types
- Case studies or real-world news headlines involving ML
- Compare traditional programming vs. ML-based approach

Activities:

- Group discussion: 'Where have you seen ML in real life?'
- Quiz on types of learning

CO2: Formulate machine learning problems and understand performance evaluation metrics.

Key Topics:

- Defining input/output, features, labels
- Concepts: Underfitting, Overfitting
- Model evaluation metrics:■ - Accuracy, Precision, Recall, F1 Score■ - Confusion Matrix■ - ROC and AUC
- Cross-validation

Teaching Aids:

- Use visuals to explain confusion matrix and ROC curves
- Hands-on with datasets (e.g., Iris dataset)
- Jupyter notebook demo for training and evaluating a simple model

Activities:

- Exercise: Evaluate two models on different metrics
- Think-pair-share: When is accuracy misleading?

CO3: Understand the basic learning algorithms - regression, classification, clustering.

Key Topics:

- Linear Regression, Logistic Regression
- k-NN, Decision Trees, Naive Bayes
- k-Means Clustering, Hierarchical Clustering

Teaching Aids:

- Visual explanations of decision boundaries
- Coding demos (Scikit-learn)
- Show clustering using color-coded plots

Activities:

- In-class model implementation (e.g., build linear regression from scratch)
- Mini-project: Choose a dataset and apply a classification algorithm

CO4: Learn feature engineering, model selection, and tuning techniques.

Key Topics:

- Data preprocessing (handling missing values, normalization)
- Feature selection and dimensionality reduction (e.g., PCA)
- Hyperparameter tuning (grid search, random search)
- Model selection strategies

Teaching Aids:

- Show impact of feature scaling
- Use Scikit-learn pipeline for preprocessing + model
- Hyperparameter tuning visualization (heatmaps)

Activities:

- Lab: Compare models with and without preprocessing
- Workshop: Feature selection on a noisy dataset

CO5: Explore modern machine learning topics: ensemble methods, neural networks.

Key Topics:

- Ensemble techniques (Bagging, Boosting, Random Forest)
- Introduction to Neural Networks
- Perceptron, Activation functions, Backpropagation (basic)
- Deep Learning overview (CNN, RNN - conceptual only)

Teaching Aids:

- Use visual tools (TensorFlow Playground)
- Ensemble method animations
- Demonstration of training a basic NN with Keras

Activities:

- Build a Random Forest on a dataset
- Class debate: 'Can machines think?'